

Exercises on Chapter 9 Uniprocessor Scheduling

Please find **10 short-term scheduling exercises**, including process, arrival time, and service time. Please practice various scheduling algorithms, such as **FCFS, RR, SPN, SRT, and HRRN**.

Assumptions for all examples:

1. Single CPU.
2. No I/O blocking.
3. Context-switching time is ignored.
4. For RR, use the quantum size for each example.
5. For SPN and SRT, "service time" is assumed to be known.
6. For ties, use FCFS order; if still tied, use alphabetical order.
7. Total time means total CPU service time, that is, the sum of all service times.

Summary of the 10 Examples

Example	No. of Processes	Total Service Time	RR Quantum	Main Focus
1	3	15	1	Simple preemption and RR comparison
2	3	20	2	Long first job with later arrivals
3	4	15	4	RR is close to FCFS for some processes
4	4	20	1	Strong RR fragmentation
5	4	25	2	SPN/SRT advantage for short jobs
6	5	20	4	Mixed short and medium processes
7	5	25	1	Good for showing SRT preemption
8	5	30	2	Longer total workload
9	6	25	4	Many processes with varied arrivals
10	6	30	1	Dense RR and SRT comparison

Example 1: 3 Processes, Total Time = 15, RR q = 1

Process	Arrival Time	Service Time
A	0	5
B	2	3
C	5	7

Example 2: 3 Processes, Total Time = 20, RR q = 2

Process	Arrival Time	Service Time
A	0	8
B	1	4
C	6	8

Example 3: 4 Processes, Total Time = 15, RR q = 4

Process	Arrival Time	Service Time
A	0	4
B	2	6
C	4	2
D	7	3

Example 4: 4 Processes, Total Time = 20, RR q = 1

Process	Arrival Time	Service Time
A	0	6
B	3	2
C	5	8
D	9	4

Example 5: 4 Processes, Total Time = 25, RR q = 2

Process	Arrival Time	Service Time
A	0	9
B	2	5
C	6	7
D	10	4

Example 6: 5 Processes, Total Time = 20, RR q = 4

Process	Arrival Time	Service Time
A	0	3
B	1	7
C	4	2
D	6	6
E	8	2

Example 7: 5 Processes, Total Time = 25, RR q = 1

Process	Arrival Time	Service Time
A	0	5
B	2	9
C	3	3
D	7	6
E	11	2

Example 8: 5 Processes, Total Time = 30, RR q = 2

Process	Arrival Time	Service Time
A	0	10
B	4	4
C	5	8
D	9	5
E	12	3

Example 9: 6 Processes, Total Time = 25, RR q = 4

Process	Arrival Time	Service Time
A	0	4
B	1	6
C	3	3
D	5	7
E	8	2
F	10	3

Example 10: 6 Processes, Total Time = 30, RR q = 1

Process	Arrival Time	Service Time
A	0	7
B	2	5
C	4	8
D	6	3
E	9	4
F	13	3

For each example, please sketch the scheduling and calculate the metrics.

Algorithm	Averaged Normalized Turnaround Time
FCFS	
RR	
SPN	
SRT	
HRRN	